Hanoi University of Science and Technology **Faculty of Mathematics and Informatics**	
Course: Calculus 2	**Course ID:** MI1121E
Academic year: 2025.2	**Training program:** Bachelor
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1.1 Week 1

Exercise 1. Find an equation of the tangent line and normal line to the curves

(a) $y = e^{1-x^2}$ at the intersection of this curve and the line $y = 1$

(b) $\begin{cases} x = 2t - \cos(\pi t) \\ y = 2t + \sin(\pi t) \end{cases}$ at the point A corresponding to $t = 1/2$

(c) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 5$ at the point $M(8; 1)$

(d) (20192-GK-4) $\begin{cases} x = 2(t - \sin t) \\ y = 2(1 - \cos t) \end{cases}$ at the point corresponding to $t = -\dfrac{\pi}{2}$.

(e) (20181-GK-1) $\begin{cases} x = (t^2 - 1)e^{2t} \\ y = (t^2 + 1)e^{3t} \end{cases}$ at the point corresponding to $t = 0$.

Exercise 2. Evaluate the curvature at the arbitrary point of

(a) $\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases}$ $(a > 0)$.

(b) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ $(a > 0)$

(c) $r = ae^{b\varphi}, (a, b > 0)$

(d) $(x - a)^2 + (y - b)^2 = R^2$, with $a, b \in \mathbb{R}, R > 0$.

(e) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, with $a, b > 0$.

Exercise 3. Evaluate the curvature of the following curves:

(a) (20192-GK-2) $\begin{cases} x = 2(t - \sin t) \\ y = 2(1 - \cos t) \end{cases}$ at the point corresponding to $t = \frac{\pi}{2}$.

(b) (20192-GK-3) $y = e^{2x}$ at the point $A(0, 1)$.

(c) (20193-GK-1) $x = \sqrt{4y} + 1$ at the point $(3, 1)$.

Exercise 4. Evaluate the curvature of the curve $y = \ln x$ at the point with positive abscissa $x > 0$. At what point does the curve have maximum curvature? If $x \to \infty$, how is the curvature?

Exercise 5. Find the envelope of the family of the following curves:

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(a) $y = \dfrac{x}{c} + c^2$

(b) $cx^2 - 3y - c^3 + 2 = 0$

(c) $y = c^2(x-c)^2$

(d) $Ax\sin c + By\cos c = C$, with $A, B \neq 0, C \in \mathbb{R}$.

(e) (20192-GK-2) $2x^2 - 4xc + 2y^2 + c^2 = 0$.

(f) (20192-GK-3) $y = 4cx^3 + c^4$.

(g) (20193-GK-1) $y = (2x+3c)^4$.

1.2 Week 2

Exercise 6. Suppose that $\vec{p}(t), \vec{q}(t), \alpha(t)$ are differentiable functions. Prove that

(a) $\dfrac{d}{dt}(\vec{p}(t) + \vec{q}(t)) = \dfrac{d\vec{p}(t)}{dt} + \dfrac{d\vec{q}(t)}{dt}$

(b) $\dfrac{d}{dt}(\alpha(t)\vec{p}(t)) = \alpha(t)\dfrac{d\vec{p}(t)}{dt} + \alpha'(t)\vec{p}(t)$

(c) $\dfrac{d}{dt}(\vec{p}(t)\vec{q}(t)) = \vec{p}(t)\dfrac{d\vec{q}(t)}{dt} + \dfrac{d\vec{p}(t)}{dt}\vec{q}(t)$

(d) $\dfrac{d}{dt}(\vec{p}(t) \times \vec{q}(t)) = \vec{p}(t) \times \dfrac{d\vec{q}(t)}{dt} + \dfrac{d\vec{p}(t)}{dt} \times \vec{q}(t)$

Exercise 7. The curve C is given by $\vec{r}(t)$. Suppose that $\vec{r}(t)$ is a differentiable function and $\vec{r}'(t)$ is always perpendicular to $\vec{r}(t)$. Prove that C lies on the sphere with center the origin.

Exercise 8. Find vector functions representing the curves given by the intersections of the following surfaces:

(a) The cylinder $x^2 + y^2 = 4$ and the surface $z = xy$.

(b) The cone $z = \sqrt{x^2 + y^2}$ and the plane $z = 1 + y$.

(c) The paraboloid $z = 4x^2 + y^2$ and the surface $y = x^2$.

(d) The upper half of the ellipsoid $x^2 + y^2 + 4z^2 = 4$, $y \geq 0$, and the cylinder $x^2 + z^2 = 1$.

Exercise 9. Let $\vec{r}(t)$ be a vector function such that $\vec{r}''(t)$ exists. Prove that

$$\frac{d}{dt}\big(\vec{r}(t) \times \vec{r}'(t)\big) = \vec{r}(t) \times \vec{r}''(t).$$

Exercise 10. Assume $\vec{r}(t) \neq \vec{0}$. Prove that

$$\frac{d}{dt}\big(|\vec{r}(t)|\big) = \frac{\vec{r}(t) \cdot \vec{r}'(t)}{|\vec{r}(t)|}.$$

Exercise 11. Find an equation of the tangent line and normal plane of the curve

(a) $\begin{cases} x = a\sin^2 t \\ y = b\sin t\cos t \\ z = c\cos^2 t \end{cases}$ at the point corresponding to $t = \dfrac{\pi}{4}$, $(a, b, c > 0)$

(b) $\begin{cases} x = 4\sin^2 t \\ y = 4\cos t \\ z = 2\sin t + 1 \end{cases}$ at $M(1; -2\sqrt{3}; 2)$

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(c) (20182-CK-5) $\begin{cases} x = t\cos 2t \\ y = t\sin 2t \\ z = 3t \end{cases}$ at the point corresponding to $t = \dfrac{\pi}{2}$.

Exercise 12. Evaluate the curvature of the curve

(a) $\begin{cases} x = \cos t \\ y = \sin t \\ z = t \end{cases}$ at the point corresponding to $t = \frac{\pi}{2}$

(b) $\begin{cases} x = \cos t + t\sin t \\ y = \sin t - t\cos t \\ z = t \end{cases}$ at the point corresponding to $t = \pi$

(c) Evaluate the curvature at the point $M(1; 0; -1)$ of the intersection curve of the cylinder $4x^2 + y^2 = 4$ and the plane $x - 3z = 4$

Exercise 13. (Υ) Let $a_i, b_i, c_i \in \mathbb{R}$ for $1 \le i \le 3$. Prove that the curve defined by the vector function

$$\vec{r}(t) = \left(a_1t^2 + b_1t + c_1,\ a_2t^2 + b_2t + c_2,\ a_3t^2 + b_3t + c_3\right)$$

is a plane curve.

Exercise 14. Find the directional derivative of the function $u = x^3 + 2y^3 + 3z^2 + 2xyz$ at $A(2; 1; 1)$ in the direction of $\vec{AB}$, where $B(3; 2; 3)$.

Exercise 15. Compute the length of vector $\overrightarrow{\text{grad}}u$, with $u = x^3 + y^3 + z^3 - 3xyz$ at $A(2; 1; 1)$. When is $\overrightarrow{\text{grad}}u$ perpendicular to Oz, and when is $\overrightarrow{\text{grad}}u = \vec{0}$?

Exercise 16. Evaulate $\overrightarrow{\text{grad}}u$, with $u = r^2 + \frac{1}{r} + \ln r$, where $r = \sqrt{x^2 + y^2 + z^2}$.

Exercise 17. Find the directions in which the rate of change of the function $u = x\sin z - y\cos z$ at the origin $O(0, 0, 0)$ is maximum?

Exercise 18. Evaluate the angle of two vectors $\overrightarrow{\text{grad}}z$ at $(3; 4)$ of the fucntions: $z = \sqrt{x^2 + y^2}$ and $z = x - 3y + \sqrt{3xy}$.

1.3 Week 3

Exercise 19. Find an equation of the tangent plane and normal line of the surface

(a) $x^2 - 4y^2 + 2z^2 = 6$ at $(2; 2; 3)$

(b) $z = 2x^2 + 4y^2$ at $(2; 1; 12)$

(c) $\ln(2x + y^2) + 3z^3 = 3$ at $(0; -1; 1)$

(d) $x^2 + 2y^3 - yz = 0$ at $(1; 1; 3)$

(e) (20152-CK-7) $z = 2x^2 + 3y^2$ at $(1, -1, 5)$.

(f) (20193-GK-1) $y^2 = 3(x^2 + z^2)$ at $(\sqrt{2}, 3, 1)$.

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Exercise 20. (20182-CK-3) Find the equation of the tangent plane to the surface $x^2 + 3y^2 - z^2 = 3$ that is parallel to the plane $x - 3y + z = 0$.

Exercise 21. Find an equation of the tangent line and normal plane of the curve

(a) $\begin{cases} x^2 + y^2 = 10 \\ y^2 + z^2 = 25 \end{cases}$ at $A(1; 3; 4)$

(b) $\begin{cases} 2x^2 + 3y^2 + z^2 = 47 \\ x^2 + 2y^2 = z \end{cases}$ at $B(-2; 1; 6)$

(c) (20182-CK-5) $\begin{cases} x^2 + y^2 + z^2 = 25 \\ 4x + 3y + 5z = 0 \end{cases}$ tại $M(3, -4, 0)$.

Exercise 22. Calculate the iterated integral by first reversing the order of integration

(a) $\displaystyle\int_{-1}^{1} dx \int_{-\sqrt{1-x^2}}^{1-x^2} f(x, y)dy$

(b) $\displaystyle\int_{0}^{1} dy \int_{2-y}^{1+\sqrt{1-y^2}} f(x, y)dx$

(c) $\displaystyle\int_{0}^{2} dx \int_{\sqrt{2x-x^2}}^{\sqrt{2x}} f(x, y)dy$

(d) $\displaystyle\int_{0}^{\frac{\pi}{2}} dy \int_{\sin y}^{1+y^2} f(x, y)dx$

(e) $\displaystyle\int_{0}^{\sqrt{2}} dy \int_{0}^{y} f(x, y)dx + \int_{\sqrt{2}}^{2} dy \int_{0}^{\sqrt{4-y^2}} f(x, y)dx$

(f) (20192-GK-3) $\displaystyle\int_{0}^{1} dy \int_{\sqrt{y}}^{\sqrt{2-y^2}} f(x, y)dx.$

Exercise 23. Calculate the value of the multiple integral

(a) $\displaystyle\iint_{\mathcal{D}} \frac{y}{1 + xy} dxdy, \mathcal{D} = \{(x, y) \in \mathbb{R}^2 : 0 \le x \le 1; 0 \le y \le 2\}$

(b) $\displaystyle\iint_{\mathcal{D}} x^2(y - x)dxdy$, where $\mathcal{D}$ is a region bounded by two curves $y = x^2$ and $x = y^2$

(c) $\displaystyle\iint_{\mathcal{D}} 2xydxdy$, where $\mathcal{D}$ is bounded by the curves $x = y^2, x = -1, y = 0$ and $y = 1$

(d) $\displaystyle\iint_{\mathcal{D}} (x + y)dxdy$, where $\mathcal{D}$ is bounded by $x^2 + y^2 \le 1, \sqrt{x} + \sqrt{y} \ge 1$

(e) $\displaystyle\iint_{\mathcal{D}} |x + y|dxdy$, where $\mathcal{D} = \{(x, y) \in \mathbb{R}^2 : |x| \le 1; |y| \le 1\}$

(f) $\displaystyle\iint_{|x|+|y|\le 1} (|x| + |y|)dxdy$

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(g) $\int\limits_0^1 dx \int\limits_0^{1-x^2} \frac{xe^{3y}}{1-y} dy$

1.4 Week 4

Exercise 24. Find the limits of integration in the polar coordinates of $\iint\limits_{\mathcal{D}} f(x,y)dxdy$, where $\mathcal{D}$ is a domain

(a) $a^2 \le x^2 + y^2 \le b^2$

(b) $x^2 + y^2 \ge 4x, x^2 + y^2 \le 8x, y \ge x, y \le \sqrt{3}x$

(c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1,\ y \ge 0, (a, b > 0)$

(d) $x^2 + y^2 \le 2x, x^2 + y^2 \le 2y$

Exercise 25. Evaluate the given integral by changing to polar coordinates

(a) $\int\limits_0^R dx \int\limits_0^{\sqrt{R^2-x^2}} \ln(1 + x^2 + y^2)dy, \quad (R > 0)$

(b) $\iint\limits_{\mathcal{D}} xydxdy$, where $\mathcal{D}$ is a half of surface: $(x-2)^2 + y^2 \le 1, y \ge 0$

(c) $\iint\limits_{\mathcal{D}} (\sin y + 3x)dxdy$, where $\mathcal{D}$ is a surface: $(x-2)^2 + y^2 \le 1$

(d) $\iint\limits_{\mathcal{D}} |x + y|dxdy$, where $\mathcal{D}$ is surface: $x^2 + y^2 \le 1$

(e) $\iint\limits_{\mathcal{D}} xy^2dxdy$, where D is a region bounded by the circles $x^2 + (y-1)^2 = 1$ and $x^2 + y^2 - 4y = 0$.

Exercise 26. Rewrite the integral $\int\limits_0^1 dx \int\limits_{-x}^{x} f(x,y)dy$ in terms of two variables u and v:

(a) Using the change of variables $u = x + y$, and $v = x - y$.

(b) Apply the result to compute the integral for $f(x,y) = (2 - x - y)^2$.

Exercise 27. Evaluate the given integrals

(a) $\iint\limits_{\mathcal{D}} \frac{2xy + 1}{\sqrt{1 + x^2 + y^2}} dxdy$, where $\mathcal{D} : x^2 + y^2 \le 1$.

(b) $\iint\limits_{\mathcal{D}} \frac{dxdy}{(x^2 + y^2)^2}$, where $\mathcal{D} : \begin{cases} y \le x^2 + y^2 \le 2y \\ x \le y \le \sqrt{3}x \end{cases}$

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(c) $\displaystyle\iint_{\mathcal{D}} \frac{xy}{x^2+y^2}dxdy$, where $\mathcal{D}: \begin{cases} 2x \le x^2+y^2 \le 12 \\ x^2+y^2 \ge 2\sqrt{3}y \\ x \ge 0, y \ge 0 \end{cases}$

(d) $\displaystyle\iint_{\mathcal{D}} |9x^2-4y^2|dxdy$, where $\mathcal{D}: \dfrac{x^2}{4}+\dfrac{y^2}{9} \le 1$.

(e) $\displaystyle\iint_{\mathcal{D}} (3x+2xy)dxdy$, where $\mathcal{D}: \begin{cases} 1 \le xy \le 9, \\ y \le x \le 4y. \end{cases}$

(f) (20152-CK-5) $\displaystyle\iint_D 3xdxdy$, where $D: \begin{cases} 0 \le x \le 2, \\ 1 \le x+y \le 3. \end{cases}$

(g) (20171-CK-1) $\displaystyle\iint_D (x-2y)dxdy$, where D bounded by the curves $y = x^2-1$ and $y = 0$.

(h) (20192-GK-3) $\displaystyle\iint_D 4ydxdy$, where $D: \begin{cases} x^2+y^2 \le 1, \\ x+y \ge 1 \end{cases}$

(i) (20152-CK-1) $\displaystyle\iint_D e^{x^2+y^2}dxdy$, where $D: \begin{cases} a^2 \le x^2+y^2 \le b^2, \\ x \ge 0, \end{cases}$ $(0 < a < b)$

Exercise 28. Compute the following double integrals:

(a) $\displaystyle\iint_D \frac{y}{x^2+1}dxdy$, where $D: \begin{cases} x \ge 0, \\ x^2+y^2 \le 1 \end{cases}$

(b) $\displaystyle\iint_D 2x(x^2+y^2)dxdy$, where $D: x^4+y^4+x^2-y^2 \le 1$

(c) $\displaystyle\iint_D |x-y|dxdy$, where $D: \begin{cases} |x| \le 1, \\ |y| \le 1 \end{cases}$

(d) $\displaystyle\iint_D xy\sqrt{x^2+y^2}dxdy$, where $D: \begin{cases} x, y \ge 0, \\ y \le x, \\ x^2+y^2 \le 1 \end{cases}$

(e) $\displaystyle\iint_D \frac{xy}{x^2+y^2}dxdy$, where $D: \begin{cases} x, y \ge 0, \\ 1 \le x^2+y^2 \le 4 \end{cases}$

(f) $\displaystyle\iint_D x^2+y^2dxdy$, where $D: \dfrac{x^2}{a^2}+\dfrac{y^2}{b^2} \le 1$, where $a, b > 0$.

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(g) $\iint_D x^3 + y^3 dxdy$, where $D : \begin{cases} x, y \geq 0, \\ \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} \leq 1, \text{ and } a, b > 0. \end{cases}$

(h) $\iint_D x^2 y dxdy$, where $D : \begin{cases} y \geq 0, \\ x^2 + y^2 - 2x \leq 0 \end{cases}$

(i) $\iint_D \dfrac{1}{(x^2+y^2)^2} dxdy$, where $D : \begin{cases} x \geq 1, \\ x^2 + y^2 - 2x \leq 0 \end{cases}$

(j) $\iint_D \sqrt{x^2+y^2} dxdy$, where $D : \begin{cases} x, y \geq 0, \\ x^2 + y^2 \leq 1, x^2 + y^2 - 2y \geq 0 \end{cases}$

(k) $\iint_D x^2 - y^2 dxdy$, where $D : (x-1)^2 + y^2 \leq 1$

(l) $\iint_D \sqrt{R^2 - x^2 - y^2} dxdy$, where $D : x^2 + y^2 - Rx \leq 0$, and $R > 0$.

(m) $\iint_D x + y dxdy$, where $D : \begin{cases} y \geq 0, \\ 0 \leq x \leq x^2 + y^2 \leq 1 \end{cases}$

(n) $\iint_D \dfrac{1}{(1+x^2+y^2)^2} dxdy$, where $D : |x| \leq x^2 + y^2 \leq 1$.

(o) $\iint_D x^2 + y^2 dxdy$, where $D : \begin{cases} y \geq 0, \\ x^2 + y^2 - x \leq 0, \\ x^2 + y^2 - y \geq 0 \end{cases}$

(p) $\iint_D \sqrt{xy} dxdy$, where $D : (x^2+y^2)^2 \leq 2xy$

(q) $\iint_D x^2 dxdy$, where $D : |r| \leq \sqrt{\cos\varphi}$

(r) $\iint_D (x+y)^2 e^{x^2-y^2} dxdy$, where $D : \begin{cases} x, y \geq 0, \\ x + y \leq 1 \end{cases}$

(s) $\iint_D (x^2 - y^2)\cos(xy) dxdy$, where $D : \begin{cases} 0 \leq x + y \leq 4, \\ x \leq y, \\ xy \geq 1 \end{cases}$

(t) $\iint_D x^2 y^2 \sqrt[3]{1 - x^3 - y^3} dxdy$, where $D : \begin{cases} x, y \geq 0, \\ x^3 + y^3 \leq 1 \end{cases}$

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(u) $\iint_D xydxdy$, where $D : \begin{cases} x, y \geq 0, \\ x^{2/3} + y^{2/3} \leq 1 \end{cases}$

(v) $\iint_D e^{\frac{x^3+y^3}{xy}} dxdy$, where $D : \begin{cases} y^2 - 2px \leq 0, \\ x^2 - 2py \leq 0, \text{ and } p > 0. \end{cases}$

(w) $\iint_D (y^2 - x^2)(x^2 + y^2)dxdy$, where $D : \begin{cases} a \leq xy \leq b, \\ y \geq x \geq 0, \\ y^2 - x^2 \leq 1, \text{ and } 0 < a < b. \end{cases}$

Exercise 29. Suppose D is the region in the half-plane $y > 0$ bounded by the four parabolas $P_1 : y^2 = 4x + 4$, $P_2 : y^2 = 2x + 1$, $P_3 : y^2 = 9 - 6x$, $P_4 : y^2 = 4 - 4x$. Compute $\iint_D \frac{y}{\sqrt{x^2 + y^2}} \, dx \, dy$ by applying the change of variables defined by: $x = u^2 - v^2,\ y = 2uv,\ u > 0,\ v > 0$.

1.5 Week 5

Exercise 30. (20192-GK-1) Tính thể tích của miền giới hạn bởi các mặt cong $y = x^2$, $x = y^2$, $z = y^2$ và mặt Oxy.

Exercise 31. Evaluate the area of the region $\mathcal{D}$ bounded by

(a) $\begin{cases} y^2 = x, y^2 = 2x \\ x^2 = y, x^2 = 2y \end{cases}$

(b) $\begin{cases} y = 0, y^2 = 4ax \\ x + y = 3a, y \leq 0, (a > 0). \end{cases}$

(c) $\begin{cases} 2x \leq x^2 + y^2 \leq 4x \\ 0 \leq y \leq x \end{cases}$

(d) $r \geq 1, r \leq \frac{2}{\sqrt{3}} \cos \varphi$

(e) $(x^2 + y^2)^2 = 2a^2xy$

(f) $r = a(1 + \cos \varphi)$ $(a > 0)$.

(g) $x^2 + y^2 + xy \leq 1$.

Exercise 32. Prove that the area of the region D defined by $x^2 + (\alpha x - y)^2 \leq 4$ is independent of α for all $\alpha \in \mathbb{R}$.

Exercise 33. Evaluate the area of a part of the sphere $x^2 + y^2 + z^2 = 4a^2$ lying inside the surface $x^2 + y^2 - 2ay = 0$, $(a > 0)$.

Exercise 34. Use a double integral to find the area of the region bounded by one petal of the four-petaled rose $r = \cos 2\varphi$.

Exercise 35. (20152-CK-7) Compute $\iint_D (x^2 + 4y^2) \, dx \, dy$, where $D : x^2 + y^2 \leq 1$.

Exercise 36. (20182-CK-3) Find the area of the region D defined by $(x^2 + y^2)^2 \leq 2x^2y,\ x \geq 0$.

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Exercise 37. Find the surface area of the curved surface:

(a)) (20192-GK-1) $z = x^2 + y^2 + 1$ inside the cylinder $x^2 + y^2 = 4$.

(b)) (20193-GK-2) $x - 2y^2 + 2z^2 = 0$ inside the cylinder $y^2 + z^2 = 1$.

1.6 Week 6

Exercise 38. Evaluate the triple integral

(a) $\iiint\limits_V zdxdydz$, where the region V is bounded by $\begin{cases} 0 \le x \le 1 \\ x \le y \le 2x \\ 0 \le z \le \sqrt{5 - x^2 - y^2} \end{cases}$

(b) $\iiint\limits_V (3xy^2 - 4xyz)dxdydz$, where the region V is bounded by $\begin{cases} 1 \le y \le 2 \\ 0 \le xy \le 2 \\ 0 \le z \le 2 \end{cases}$

(c) $\iiint\limits_V xye^{yz^2}dxdydz$, where the region V is bounded by $\begin{cases} 0 \le x \le 1 \\ 0 \le y \le 1 \\ x^2 \le z \le 1 \end{cases}$

(d) $\iiint\limits_V (x^2 + y^2)dxdydz$, where the region V is bounded by $\begin{cases} x^2 + y^2 + z^2 \le 1 \\ x^2 + y^2 - z^2 \le 0 \end{cases}$

(e) (20152-CK-7) $\iiint_V yzdxdydz$, trong đó miền $V : \begin{cases} z^2 \le x \le \sqrt{z}, \\ 0 \le y \le z, \\ 0 \le z \le 1. \end{cases}$

Exercise 39. $\iiint\limits_V z\sqrt{x^2 + y^2}dxdydz$, where

(a) V is bounded by the cylinder $x^2 + y^2 = 2x$ and planes: $y = 0, z = 0, z = a$, $(y \ge 0, a > 0)$

(b) V is a half of the sphere $x^2 + y^2 + z^2 \le a^2, z \ge 0, (a > 0)$

(c) V is a half of the elipsoid $\frac{x^2+y^2}{a^2} + \frac{z^2}{b^2} \le 1, z \ge 0, (a, b > 0)$

Exercise 40. Compute $\iiint\limits_V ydxdydz$, where V is a region bounded by the cone: $y = \sqrt{x^2 + z^2}$ and the plane $y = h, (h > 0)$

Exercise 41. Evaluate the triple integral

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(a) $\iiint\limits_V \frac{x^2}{a^2}dxdydz$, where $V: \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\le 1\ (a,b,c>0)$

(b) $\iiint\limits_V (x^2+y^2+z^2)dxdydz$, where $V: \begin{cases} 1\le x^2+y^2+z^2\le 4 \\ x^2+y^2\le z^2 \end{cases}$

(c) $\iiint\limits_V \sqrt{x^2+y^2}dxdydz$, where V is a region bounded by $x^2+y^2=z^2, z=-1$

(d) $\iiint\limits_V \frac{dxdydz}{[x^2+y^2+(z-2)^2]^2}$, where $V: \begin{cases} x^2+y^2\le 1 \\ |z|\le 1 \end{cases}$

(e) $\iiint\limits_V \sqrt{x^2+y^2+z^2}dxdydz$, where V is the region bounded by $x^2+y^2+z^2\le z$

Exercise 42. Evaluate the triple integral

(a) $\iiint_D x^2ye^{xyz}dxdydz$, where $D=[0,1]^3$.

(b) $\iiint_D zdxdydz$, where $D: x,y,z\ge 0, \sqrt{x}+\sqrt{y}+\sqrt{z}\le 1$.

(c) $\iiint_D z^2dxdydz$, where $D: x^2+y^2\le 1, 0\le z\le 1$.

(d) $\iiint_D x^2zdxdydz$, where $D: x^2+y^2\le z^2, 0\le z\le 1$.

(e) $\iiint_D \frac{1}{\sqrt{x^2+y^2+(z-a)^2}}dxdydz$, where $D: x^2+y^2+z^2\le R^2 \quad (a>R>0)$.

(f) $\iiint_D \frac{1}{(x^2+y^2+1)^2}dxdydz$, where $D: x^2+y^2-y\le 0,\ 0\le z\le 1$.

(g) $\iiint_D x^2+y^2dxdydz$, where $D: x^2+y^2+z^2\le 1$.

(h) $\iiint_D (x+y+z)^2dxdydz$, where $D: x^2+y^2+z^2\le R^2,\ (R>0)$.

(i) $\iiint_D x^2+y^2+z^2dxdydz$, where $D: 1\le \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\le 4,\ (a,b,c>0)$.

(j) $\iiint_D xyzdxdydz$, where $D: \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\le 1, x,y,z>0,\ (a,b,c>0)$..

(k) $\iiint_D (3x+2y+z)^2dxdydz$, where $D: 9x^2+4y^2+z^2\le 1$.

(l) $\iiint_D \sqrt{1+\sqrt{x^2+y^2+z^2}}dxdydz$, where $D: x^2+y^2+z^2\le 1$.

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(m) $\iiint_D \frac{1-(x^2+y^2+z^2)}{1+(x^2+y^2+z^2)}dxdydz$, where $D: x^2+y^2+z^2 \leq 1$.

(n) $\iiint_D zdxdydz$, where $D: x^2+y^2 \leq z^2, x^2+y^2+z^2 \leq 1, z \geq 0$.

(o) (Υ) $\iiint_D \sqrt{xyz}dxdydz$, where $D: x, y, z \geq 0, x+y+z \leq 1$.

(p) (Υ) $\iiint_D \frac{1}{(1+y+z)^2}dxdydz$, where $D: x, y, z \geq 0, x+y+z \leq 1$.

(q) (Υ) $\iiint_D \sqrt{xyz}dxdydz$, where $D: x, y, z \geq 0, ax+by+cz \leq d$, $(a, b, c, d > 0)$.

1.7 Week 7

Exercise 43. Evaluate the volume of the region given by

(a) $\begin{cases} x+y \geq 1 \\ x+2y \leq 2 \\ y \geq 0, 0 \leq z \leq 2-x-y \end{cases}$

(b) $|x-y|+|x+3y|+|x+y+z| \leq 1$.

Exercise 44. Evaluate the volume of the region is bounded by the surfaces

(a) $\begin{cases} z = 4-x^2-y^2 \\ 2z = 2+x^2+y^2 \end{cases}$

(b) $z = 1+x^2+y^2$, surface $x^2+4y^2 = 4$ and the plane Oxy.

(c) $az = x^2+y^2, z = \sqrt{x^2+y^2}, (a > 0)$.

Exercise 45. On which intervals the following function $I(y) = \int_0^1 \frac{y^2-x^2}{(x^2+y^2)^2}dx$ is continuous.

Exercise 46. Let $f(x)$ is a positive function, and continuous on $[0, 1]$. Prove that the integral $I(y) = \int_0^1 \frac{yf(x)}{x^2+y^2}dx$ is not continuous at 0.

1.8 Week 8

Exercise 47. Given a function $f(y) = \int_0^{\frac{\pi}{2}} \ln(\sin^2 x + y^2\cos^2 x)dx$. Evaluate $f'(1)$.

Exercise 48. Evaluate

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(a) $\lim\limits_{y\to 1}\int\limits_0^y \frac{\arctan x}{x^2+y^2}dx.$

(b) $\lim\limits_{y\to 0}\int_0^1 (x+3y)\sqrt{x^2+y^3}dx$

(c) $\lim\limits_{y\to 1}\int_y^{\ln y}(x+y)dx$

(d) $\lim\limits_{y\to 0}\int\limits_y^{1+y}\frac{dx}{1+x^2+y^2}$

(e) $\lim\limits_{y\to 0}\int\limits_0^2 x^2\cos xydx$

Exercise 49. Evaluate the following integrals:

(a) $I_n(\alpha)=\int_0^1 x^\alpha \ln^n x dx, \quad (n\in\mathbb{N})$

(b) (GK-2016) $\int_0^{+\infty}\frac{e^{-ax^2}-1}{x^2e^{x^2}}dx$, với $a>0$.

Exercise 50. Investigate the continuity and differentiability of the function $g(y)=\int_0^1\frac{dx}{x^2+y^2}$.

Exercise 51. Prove that the parameter-dependent integral $I(y)=\int\limits_{-\infty}^{+\infty}\frac{\arctan(x+y)}{1+x^2}dx$ defines a function that is continuous and differentiable with respect to y. Compute $I'(y)$ and hence determine an explicit expression for $I(y)$.

Exercise 52. Evaluate

(a) $I(y)=\int\limits_0^1\arctan\frac{x}{y}dx$

(b) $J(y)=\int\limits_0^1\ln(x^2+y^2)dx$

(c) $K=\int\limits_0^1\frac{x^b-x^a}{\ln x}dx$, $0<a<b$.

Exercise 53. Prove that

(a) $\int\limits_0^{+\infty}e^{-x^2}dx=\frac{\sqrt{\pi}}{2}$

(b) $\int\limits_0^{+\infty}e^{-yx^2}dx=\frac{\sqrt{\pi}}{2\sqrt{y}}, \quad y>0$

(c) $\int\limits_0^{+\infty}\frac{\sin x}{x}e^{-yx}dx=\frac{\pi}{2}-\arctan y$

(d) $\int\limits_0^{+\infty}\frac{\sin x}{x}dx=\frac{\pi}{2}$

(e) $\int\limits_0^{+\infty}\sin(x^2)dx=\int\limits_0^{+\infty}\cos(x^2)dx=\frac{\sqrt{2\pi}}{4}$

(f) $\int\limits_0^{+\infty}\frac{\sin yx}{x(1+x^2)}dx=\frac{\pi}{2}(1-e^{-y}), \quad y\geq 0$

(g) $\int\limits_0^{+\infty}\frac{1-\cos yx}{x^2}dx=\frac{|y|\pi}{2}$

(h) $\int\limits_0^{+\infty}\frac{x\sin yx}{a^2+x^2}dx=\frac{\pi}{2}e^{-ay}, \quad a,y\geq 0.$

(i) $\int\limits_0^{+\infty}\left(e^{-\frac{a}{x^2}}-e^{-\frac{b}{x^2}}\right)dx=\sqrt{\pi b}-\sqrt{\pi a}$, $(a,b>0)$.

(j) $\int\limits_0^{+\infty}\frac{\arctan\frac{x}{a}-\arctan\frac{x}{b}}{x}dx=\frac{\pi}{2}\ln\frac{b}{a}$, $(a,b>0)$.

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Exercise 54. Show that the integral

(a) $I(y) = \int\limits_1^\infty \sin(yx)dx$ is convergent if $y = 0$ and is divergent if $y \neq 0$.

(b) $I(y) = \int\limits_0^\infty \frac{\cos \alpha x}{x^2+1}dx$ is uniformly convergent on $\mathbb{R}$.

(c) $I(y) = \int\limits_0^1 x^{-y}dx = \int\limits_1^\infty t^{y-2}dt$ is convergent if $y < 1$ and divergent if $y \geq 1$.

(d) $I(y) = \int\limits_0^{+\infty} e^{-yx} \cdot \frac{\sin x}{x}dx$ is uniformly convergent on $[0, +\infty)$.

Exercise 55. (a) Evaluate $I(y) = \int\limits_0^{+\infty} ye^{-yx}dx$, where $y > 0$.

(b) Prove that $I(y)$ uniformly converges to 1 on $[y_0, +\infty)$ for all $y_0 > 0$.

(c) Explain why $I(y)$ is not uniformly convergent on $(0; +\infty)$.

Exercise 56. Explain why

$$\lim_{y\to 0^+}\left(\int\limits_0^{+\infty} ye^{-yx}dx\right) \neq \int\limits_0^{+\infty}\left(\lim_{y\to 0^+} ye^{-yx}\right)dx.$$

1.9 Week 9

Exercise 57. Evaluate the following integrals (with $a, b, \alpha, \beta > 0$ and $n \in \mathbb{Z}^+$):

(a) $\int\limits_0^1 \frac{x^b - x^a}{\ln x}dx$

(b) $\int\limits_0^\infty \frac{e^{-\alpha x} - e^{-\beta x}}{x}dx$

(c) $\int\limits_0^{+\infty} e^{-ax}\frac{\sin(bx) - \sin(cx)}{x}dx$

(d) $\int\limits_0^1 x^\alpha (\ln x)^n dx$

(e) $\int\limits_0^{+\infty} \frac{dx}{(x^2+y)^{n+1}}$

(f) $\int\limits_0^{\frac{\pi}{2}} \ln(1 + y\sin^2 x)dx$, with $y > -1$

(g) $\int\limits_0^{+\infty} \frac{\sin xy}{x}dx, \quad y \geq 0$

(h) $\int\limits_0^{+\infty} x^{2n}e^{-x^2}\cos bx dx, \quad n \in \mathbb{N}$

(i) $\int\limits_0^{+\infty} \frac{\sin ax \cos bx}{x}dx$

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Exercise 58. Evaluate $(a, b, \alpha, \beta > 0)$:

(a) $\displaystyle\int_0^{\frac{\pi}{2}} \sin^6 x \cos^4 x dx$

(b) $\displaystyle\int_1^{+\infty} \frac{(\ln x)^4}{x^2} dx$

(c) $\displaystyle\int_0^{+\infty} x^{10} e^{-x^2} dx$

(d) $\displaystyle\int_0^{+\infty} \frac{\sqrt{x}}{(1+x^2)^2} dx$

(e) $\displaystyle\int_0^{+\infty} \frac{1}{1+x^3} dx$

(f) $\displaystyle\int_0^{+\infty} \frac{x^{n+1}}{(1+x^n)^2} dx, (2 < n \in \mathbb{N})$

(g) $\displaystyle\int_{-\infty}^{0} e^{2x} \sqrt[3]{1-e^{3x}} dx$

(h) $\displaystyle\int_0^{a} x^{2n} \sqrt{a^2-x^2} dx, \ (a > 0, n \in \mathbb{N})$

(i) $\displaystyle\int_0^{+\infty} \frac{x^4}{(1+x^3)^2} dx$

(j) $\displaystyle\int_0^{1} \frac{1}{\sqrt[n]{1-x^n}} dx, (2 \leq n \in \mathbb{N})$

(k) $\displaystyle\int_0^{+\infty} \frac{e^{-\alpha x^2} - e^{-\beta x^2}}{x^2} dx$

(l) $\displaystyle\int_0^{+\infty} \frac{\cos bx - \cos cx}{x} e^{-ax} dx$

(m) $\displaystyle\int_0^{+\infty} e^{-ax} \cos(yx) dx$

(n) $\displaystyle\int_0^{+\infty} e^{-x^2} \cos(yx) dx$

(o) $\displaystyle\int_{-\infty}^{+\infty} \frac{\arctan(x+y)}{1+x^2} dx$

(p) $\displaystyle\int_0^{+\infty} \frac{e^{-ax^2} - e^{-bx^2}}{x} dx$

(q) $\displaystyle\int_0^{+\infty} \frac{e^{-ax^3} - e^{-bx^3}}{x} dx$

(r) $\displaystyle\int_0^{+\infty} \frac{e^{-ax^2} - \cos bx}{x^2} dx$

(s) $\displaystyle\int_0^{\pi} \ln(1 + y \cos x) dx$

(t) $\displaystyle\int_0^{+\infty} e^{-x^2} \sin ax dx$

(u) (CK2014) $\displaystyle\int_0^{+\infty} \frac{x^2 dx}{(4+x^2)^3} dx$

(v) (CK2014) $\displaystyle\int_0^{+\infty} x^5 2^{-x^2} dx$

(w) (CK2014) $\displaystyle\int_0^{1} \frac{x^3(x^a - 1)}{\ln x} dx \ (a > -3)$

(x) (CK2015) $\displaystyle\int_0^{+\infty} x^{17} e^{-x^2} dx$

(y) (CK2015) $\displaystyle\int_0^{1} \sqrt[4]{\frac{x}{(1-\sqrt{x})^2}} dx$

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1.10 Week 10

Exercise 59. Evaluate the following line integrals

(a) $\int\limits_C (3x-y)ds$, where C is a half of the circle $y=\sqrt{9-x^2}$

(b) $\int\limits_C (x-y)ds$, where C is a circle $x^2+y^2=2x$

(c) $\int\limits_C y^2 ds$, where C is a curve $\begin{cases} x=a(t-\sin t) \\ y=a(1-\cos t) \end{cases}$ $(0\le t\le 2\pi, a>0)$

(d) $\int\limits_C \sqrt{x^2+y^2}ds$, where C is a curve $\begin{cases} x=a(\cos t+t\sin t) \\ y=a(\sin t-t\cos t) \end{cases}$ $(0\le t\le 2\pi, a>0)$

Exercise 60. Evaluate the following line integrals

(a) $\int\limits_{AB} (x^2-2xy)dx+(2xy-y^2)dy$, where AB is a part of parabol $y=x^2$ from $A(1;1)$ to $B(2;4)$.

(b) $\int\limits_C (2x-y)dx+xdy$, where C is a curve $\begin{cases} x=a(t-\sin t) \\ y=a(1-\cos t) \end{cases}$ whose direction is increasing direction of the parameter t, $(0\le t\le 2\pi, a>0)$.

(c) $\int\limits_{ABCA} 2(x^2+y^2)dx+x(4y+3)dy$, where $ABCA$ is a broken line through the points $A(0;0)$, $B(1;1)$, $C(0;2)$

(d) $\int_C x^2\,dy+y^2\,dx$, where C is the upper half of the ellipse $x^2+4y^2-4=0$, $y\ge 0$, traversed in the clockwise direction.

(e) $\int_C y\sin x\,dx+x\cos y\,dy$, where C is the line segment OA from $O(0,0)$ to $A(1,1)$.

(f) $\int_C \frac{x-y}{x^2+y^2}\,dx+\frac{x+y}{x^2+y^2}\,dy$, where C is the boundary of the square $ABCD$, with $A(1,1)$, $B(-1,1)$, $C(-1,-1)$, $D(1,-1)$.

(g) $\int_C x^2\,dy+y^2\,dx$, where C is the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$, traversed once counterclockwise.

(h) $\int_C xy^2\,dy-yx^2\,dx$, where C is the circle $x^2+y^2-2y=0$, traversed once counterclockwise.

(i) $\int_C y\,dx+2x\,dy$, where C is the boundary of the region defined by $x^2+y^2-2x\le 0$ and $x^2+y^2-2y\le 0$, traversed once counterclockwise.

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(j) $\displaystyle\int_C \frac{xy}{x^2+y^2}(y\,dx - x\,dy)$, where $C : \begin{cases} x = a\cos t\sqrt{\cos 2t} \\ y = a\sin t\sqrt{\cos 2t} \end{cases}$, with t varying from $-\frac{\pi}{4}$ to $\frac{\pi}{4}$ $(a > 0)$.

(k) $\displaystyle\int_C y\,dx - z\,dy + x\,dz$, where $C : \begin{cases} x = \cos t \\ y = \cos t \\ z = \sin t \end{cases}$, with t varying from 0 to 2π.

(l) $\displaystyle\int_C \frac{(y+z)\,dx + (z+x)\,dy + z\,dz}{x^2+y^2}$, where $C : \begin{cases} x = r\cos t \\ y = r\sin t \\ z = ht, \end{cases}$ with t varying from 0 to 2π, $(r > 0, h \neq 0)$.

(m) $\displaystyle\int_C x^2\,dy + y^2\,dz + z^2\,dx$, where $C : \begin{cases} x = \cos t + 1 \\ y = \cos t - 1 \\ z = \sin t \end{cases}$, with t varying from 0 to $\frac{\pi}{2}$.

Exercise 61. Does there exist a vector field $\vec{F}$ such that $\operatorname{curl}\vec{F} = \nabla \times \vec{F} = 2x\vec{i} + 3yz\vec{j} - xz^2\vec{k}$?

Exercise 62. Evaluate $\displaystyle\int_C \vec{F}\cdot d\vec{r}$, where

$$\vec{F} = e^y\vec{i} + (xe^y + e^z)\vec{j} + ye^z\vec{k}$$

and C is the line segment from $(0,2,0)$ to $(4,0,3)$.

1.11 Week 11

Exercise 63. Compute the following line integrals:

(a) $\displaystyle\int_C (x^2 - y)\,dx + (y^2 - x)\,dy$, where C is the loop (closed arc) of the curve given by $x^3 + y^3 - 3xy = 0$, traversed once counterclockwise.

(b) $\displaystyle\int_C 2xy^2z\,dx + 2x^2yz\,dy + (x^2y^2 - 2z)\,dz$, where $C : x = \cos t,\ y = \frac{\sqrt{3}}{2}\sin t,\ z = \frac{1}{2}\sin t,\ t \in [0, 2\pi]$.

(c) $\displaystyle\int_C y\,dx + x\,dy$, where C is the parabolic arc $y = x^2$ from $O(0,0)$ to $A(2,4)$.

(d) $\displaystyle\int_C \frac{-2xy\,dx + (x^2+y^2)\,dy}{x^2+y^2}$, where C is the quarter-circle arc of $x^2+y^2-y = 0$ from $A\left(\frac{1}{2}, \frac{1}{2}\right)$ to $B(0,1)$.

(e) $\displaystyle\int_C (y^3 - 6xy^2)\,dx + (3xy^2 - 6x^2y)\,dy$, where C is the semicircle with diameter AB, from $A(1,2)$ to $B(3,4)$ lying above AB.

(f) $\displaystyle\int_C (x^2+y)\,dx + (2x+y^2)\,dy$, where C is the square $ABCD$, with $A(1,1)$, $B(2,1)$, $C(2,2)$, and $D(1,2)$, traversed once counterclockwise.

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(g) $\int_C (x^2+y^2)\,dx + (x^2-y^2)\,dy$, where C is the triangle OAB, with $A(1,0)$ and $B(0,1)$, traversed once counterclockwise.

(h) $\int_C 2(x^2+y^2)\,dx + (x+y)^2\,dy$, where C is the triangle ABC, with $A(1,1)$, $B(2,2)$, and $C(1,3)$, traversed once counterclockwise.

(i) $\int_C (e^x\cos y + xy^2)\,dx - (e^x\sin y + x^2y)\,dy$, where C is a portion of the lemniscate given in polar coordinates by $r=\sqrt{\cos 2\varphi}$, with φ varying from 0 to $\frac{\pi}{4}$.

1.12 Week 12

Exercise 64. Evaluate the following line integral

$$\int_C (xy+x+y)dx + (xy+x-y)dy$$

in two ways: by computing it directly, and by Green's formula, then compare the results, where C is a curve:

(a) $x^2+y^2=R^2$ (b) $x^2+y^2=2x$ (c) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, (a,b>0)$

Exercise 65. Evaluate the following line integrals

(a) $\displaystyle\int_{ABCDA} \frac{dx+dy}{|x|+|y|}$, where $ABCDA$ is a broken line through the points $A(1;0)$, $B(0;1)$, $C(-1;0)$, $D(0;-1)$

(b) $\displaystyle\oint_{x^2+y^2=2x} x^2(y+\frac{x}{4})dy - y^2(x+\frac{y}{4})dx$

(c) $\displaystyle\oint_{OABO} e^x[(1-\cos y)dx-(y-\sin y)dy]$, where $OABO$ is a broken line through the points $O(0;0)$, $A(1;1)$, $B(0;2)$

(d) $\displaystyle\oint_{x^2+y^2=2x} (xy+e^x\sin x+x+y)dx - (xy-e^{-y}+x-\sin y)dy$

(e) $\displaystyle\oint_C (xy^4+x^2+y\cos(xy))dx + (\frac{x^3}{3}+xy^2-x+x\cos(xy))dy$, where C is a circle $\begin{cases} x=a\cos t \\ y=a\sin t, \end{cases}$ $(a>0)$

Exercise 66. Using the line integral of the second kind in order to compute the area of the region bounded by an arch of the cycloid: $x=a(t-\sin t), y=a(1-\cos t)$ and x-axis, $(a>0)$.

Exercise 67 (Υ)**.** Compute the areas of the following regions D (improper integrals may be required):

(a) The region D bounded by the loop of the curve $\begin{cases} x=t^2+t^3, \\ y=t^2+t^3-2t^4-2t^5 \end{cases}$.

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(b) The region D bounded by the curve $\begin{cases} x = \cos^2 t, \\ y = \cos t(1+\sin t) \end{cases}$.

(c) The region D is the loop of the strophoid $r = \dfrac{\cos 2\varphi}{\cos\varphi}$.

(d) The region D is the loop of $r = \sqrt{\sin^3\varphi}$.

(e) The region D bounded by $r = \dfrac{1}{2+\cos 4\varphi}$.

(f) The region D bounded by $r = \cos^3\varphi - \sin^3\varphi$.

(g) The region D is the loop of $r = \dfrac{\cos\varphi}{1-\cos\varphi}$.

(h) The region D is the loop of $r = 1 + \tan\dfrac{\varphi}{2}$.

(i) The region D bounded by the cardioid $r = 2\cos\varphi - 1$.

Exercise 68. Compute the following line integrals:

(a) $\displaystyle\oint_C \frac{(x+y)\,dy + (x-y)\,dx}{x^2+y^2}$, where $C : x^2+y^2 = 4$, traversed counterclockwise.

(b) $\displaystyle\oint_{C_\alpha} \frac{(x-y)\,dx + (x+y)\,dy}{x^2+y^2}$, where C_α is the curve $x^2+\alpha y^2 = 1$, with positive orientation, $\alpha > 0$.

(c) $\displaystyle\oint_C \frac{xy}{x^2+y^2}(y\,dx - x\,dy)$, where $C : x = \cos t\sqrt{\cos 2t}, \quad y = \sin t\sqrt{\cos 2t}$, with t varying from $-\frac{\pi}{4}$ to $\frac{\pi}{4}$.

Exercise 69. Evaluate the following line integrals

(a) $\displaystyle\int_{(-2;-1)}^{(3;0)} (x^4+4xy^3)dx + (6x^2y^2-5y^4)dy$ (b) $\displaystyle\int_{(1;\pi)}^{(2;2\pi)} (1-\frac{y^2}{x^2}\cos\frac{y}{x})dx + (\sin\frac{y}{x} + \frac{y}{x}\cos\frac{y}{x})dy$

Exercise 70. Evaluate the line integral $I = \displaystyle\int_L (3x^2y^2 + \frac{2}{4x^2+1})dx + (3x^3y + \frac{2}{y^3+4})dy$ where L is a curve $y = \sqrt{1-x^4}$ from $A(1;0)$ to $B(-1;0)$.

Exercise 71. Find the constant α such that the following integral is an independent of path in the domain

$$\int_{AB} \frac{(1-y^2)dx + (1-x^2)dy}{(1+xy)^\alpha}.$$

Exercise 72. Find the constants a, b such that $(y^2+axy+y\sin(xy))dx + (x^2+bxy+x\sin(xy))dy$ is total differetial of the function $u(x,y)$. Find the function $u(x,y)$.

Exercise 73. Find the function $h(x)$ such that the integral

$$\int_{AB} h(x)[(1+xy)dx + (xy+x^2)dy]$$

is independent of the path in the domain. With the function $h(x)$ found, evaluate the integral above from $A(2;0)$ to $B(1;2)$.

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Exercise 74. Find the function $h(xy)$ in order to the integral

$$\int\limits_{AB} h(xy)[(y + x^3y^2)dx + (x + x^2y^3)dy]$$

is not dependent on the path in the domain. With the function $h(xy)$ just found, find the above integral from $A(1;1)$ to $B(2;3)$.

Exercise 75. Compute the line integral $\displaystyle\int_C \arctan\frac{x}{y}\,(x\,dx + y\,dy)$, where C is the curve given by $x = \sqrt{3} + \sqrt{2}\cos t$ and $y = 3 + \sqrt{2}\sin t$, with t ranging from $-\frac{3\pi}{4}$ to $\frac{\pi}{4}$.

Exercise 76. Compute $\displaystyle\int_L e^{x+y^2}\left[(x^2+2x)\,dx + 2x^2y\,dy\right]$, where L is the parabola $y = 1 - x^2$ from $A(1,0)$ to $B(0,1)$.

1.13 Week 13

Exercise 77. Evaluate the following surface integrals

(a) $\displaystyle\iint\limits_S (z + 2x + \frac{4y}{3})dS$, where $S = \{(x,y,z) : \frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1, x \geq 0, y \geq 0, z \geq 0\}$

(b) $\displaystyle\iint\limits_S (x^2 + y^2)dS$, where $S = \{(x,y,z) : z = x^2 + y^2; 0 \leq z \leq 1\}$

(c) $\displaystyle\iint\limits_S z dS$, where $S = \{(x,y,z) : y = x + z^2, 0 \leq x \leq 1, 0 \leq z \leq 1\}$

(d) $\displaystyle\iint\limits_S \frac{dS}{(1+x+y+z)^2}$, where S is a boundary of the tetrahedron $x + y + z \leq 2, x \geq 0, y \geq 0, z \geq 0$

(e) $\displaystyle\iint\limits_S z(x^2+y^2)\,dx\,dy$, where S is the upper hemisphere $x^2+y^2+z^2 = 1$, $z \geq 0$, with outward orientation.

(f) $\displaystyle\iint\limits_S y\,dz\,dx + z^2\,dx\,dy$, where S is the outer surface of the ellipsoid $x^2 + \frac{y^2}{4} + z^2 = 1$, with $x \geq 0,\ y \geq 0,\ z \geq 0$.

(g) $\displaystyle\iint\limits_S x^2y^2z\,dx\,dy$, where S is the lower hemisphere $x^2 + y^2 + z^2 = R^2$, $z \leq 0$.

(h) $\displaystyle\iint\limits_S (y + z)\,dx\,dy$, where S is the part of the surface $z = 4 - 4x^2 - y^2$ lying above the plane $z \geq 0$.

(i) (20171-CK-5) $\displaystyle\iint_S z^2\sqrt{2x - x^2 - y^2}\,dx\,dy$, where S is the surface $z = \sqrt{2x - x^2 - y^2}$ oriented in the direction $z > 0$.

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(j) (20161-CK-1) $\iint_S z\,dx\,dy$, where S is the upper hemisphere $x^2+y^2+z^2=9$, $z\geq 0$, with outward orientation.

(k) (20152-CK-7) $\iint_S (x+z)\,dx\,dy$, where S is the part of the surface $z=x^2+4y^2$ with $z\leq 4$.

1.14 Week 14

Exercise 78. Evaluate the following surface integrals

(a) $\iint\limits_S x^3\,dy\,dz+y^3\,dz\,dx+z^3\,dx\,dy$, where S is the outer surface of the sphere $x^2+y^2+z^2=R^2$.

(b) $\iint\limits_S y^2z\,dx\,dy+xz\,dy\,dz+x^2y\,dz\,dx$, where S is the boundary (with outward orientation) of the region $x^2+y^2\leq 1, 0\leq z\leq x^2+y^2, x\geq 0,\ y\geq 0$.

(c) $\iint\limits_S x\,dy\,dz+y\,dz\,dx+z\,dx\,dy$, where S is the boundary (with outward orientation) of the region $(z-1)^2\geq x^2+y^2, a\leq z\leq 1$.

(d) (20151-CK) $\iint_S (x^2+y^2)\,dy\,dz+(x+3z^2)\,dz\,dx+4xz\,dx\,dy$, where the closed surface S is the boundary of the region $V: x+y+z\leq 1$, $x\geq 0$, $y\geq 0$, $z\geq 0$, with outward orientation.

(e) (20182-CK-1) $\iint_S y\,dz\,dx+z\,dx\,dy$, where S is the lower side of the cone $z=\sqrt{x^2+y^2}$, $0\leq z\leq 1$, viewed in the positive direction of the Oz-axis.

(f) (20162-CK) $\iint_S (x^3+2yz)\,dy\,dz+(3x^2y+y)\,dz\,dx+(6y^2z+xy)\,dx\,dy$, where S is the paraboloid $z=x^2+y^2$ with $z\leq 1$, oriented downward.

(g) (20162-CK) $\iint_S x\,dy\,dz$, where S is the surface $x=y^2+3z^2$, $x\leq 3$, oriented in the positive direction of the Ox-axis.

1.15 Week 15

Exercise 79. Use Stokes Theorem to evaluate the line integral $\int\limits_C (x+y^2)dx+(y+z^2)dy+(z+x^2)dz$, where C is a boundary of the triangle $(1;0;0),(0;1;0),(0;0;1)$, oriented counterclockwise when viewed from above.

Exercise 80. Given the sphere S: $x^2+y^2+z^2=1$ lies inside the cylinder $x^2+x+z^2=0, y\geq 0$, which is oriented outward. Prove that: $\iint\limits_S (x-y)dxdy+(y-z)dydz+(z-x)dzdx=0$.

Exercise 81. Evaluate

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(a) $\int_C y\,dz + z\,dx + x\,dy$, where C is the circle $\begin{cases} x^2+y^2+z^2=1, \\ x+y=1 \end{cases}$, traversed once counterclockwise as determined by the normal vector $\vec{u}=(1,1,0)$.

(b) $\int_C (y+z)\,dx + (z+x)\,dy + (x+y)\,dz$, where C is the circle $\begin{cases} x^2+y^2+z^2=1, \\ x+y+z=0 \end{cases}$, traversed once counterclockwise as determined by the normal vector $\vec{u}(1,1,1)$.

Exercise 82. Let C be a simple closed smooth curve that lie in the plane $x+y+z=1$. Show that the line integral $\int_C z\,dx - 2x\,dy + 3y\,dz$ depends only on the area of the region enclosed by C not on the shape of C or its location in the plane.

Exercise 83. Evaluate $\int_C (y+\sin x)\,dx + (z^2+\cos y)\,dy + x^3\,dz$, where C is the curve $x=\sin t, y=\cos t, z=\sin 2t$, t increases from 0 to 2π.

Exercise 84. Evaluate $\iint_S (2x+2y+z^2)\,dS$, where S is the sphere $x^2+y^2+z^2=1$.

Exercise 85. Evaluate $\iint_S \left(\frac{1}{x}\,dy\,dz + \frac{1}{y}\,dz\,dx + \frac{1}{z}\,dx\,dy\right)$, where S is the surface $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$, with outward orientation.

Exercise 86. Evaluate $\iint_S \frac{dS}{\sqrt{\frac{x^2}{a^4}+\frac{y^2}{b^4}+\frac{z^2}{c^4}}}$, where S is the surface $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$.

Exercise 87. Evaluate $\iint_S \frac{x\,dy\,dz + y\,dz\,dx + z\,dx\,dy}{(x^2+y^2+z^2)^{3/2}}$, where S: $4x^2+9y^2+6z^2=36$.

1.16 Week 16

Exercise 88. Which of the following fields are conservative and find their potential functions.

(a) $\vec{F}=5(x^2-4xy)\vec{i}+(3x^2-2y)\vec{j}+\vec{k}$

(b) $\vec{F}=(yz-3x^2)\vec{i}+xz\vec{j}+(xy+2)\vec{k}$

(c) $\vec{F}=(x+y)\vec{i}+(x+z)\vec{j}+(z+y)\vec{k}$

(d) $\vec{F}=C\frac{x\vec{i}+y\vec{j}+z\vec{k}}{\sqrt{(x^2+y^2+z^2)^3}}, C\neq 0$ is a constant

(e) $\vec{F}=(\arctan z+4xyz)\vec{i}+(2x^2z-3y^2)\vec{j}+(\frac{x}{1+z^2}+2x^2y)\vec{k}$

Exercise 89. Compute the flux of the vector field $\vec{F}$ across the following surfaces S:

(a) $\begin{cases} \vec{F}(x,y,z)=(0,0,z), \\ S: x^2+y^2+z^2=1, \text{ outward orientation.} \end{cases}$

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(b) $\begin{cases} \vec{F}(x,y,z) = (y, x, x+y), \\ S \text{ is the triangle } x+2y+z=2,\ x \geq 0,\ y \geq 0,\ z \geq 0, \text{ oriented upward.} \end{cases}$

(c) $\begin{cases} \vec{F}(x,y,z) = (y^2, x^2, z), \\ S \text{ is the part of the cylinder given in cylindrical coordinates by } x = a\cos\varphi,\ y = a\sin\varphi,\ 0 \leq z \leq h, \\ 0 \leq \varphi \leq \dfrac{\pi}{2},\ (a, h > 0), \text{ oriented outward from the cylinder.} \end{cases}$

(d) $\begin{cases} \vec{F} = xz^2\vec{i} + yx^2\vec{j} + zy^2\vec{k}, \\ S : x^2+y^2+z^2 = 1, \text{ outward orientation.} \end{cases}$

(e) $\begin{cases} \vec{F} = (x+y)z^2\vec{i} + (y+z)x^2\vec{j} + (z+x)y^2\vec{k}, \\ S : x^2+y^2+z^2 = 1, \text{ outward orientation.} \end{cases}$

(f) $\begin{cases} \vec{F}(x,y,z) = (1-x^2)\vec{i} + \frac{y^2}{2}\vec{j} + z(2x-y)\vec{k}, \\ S \text{ is the outer surface of the cylinder } x^2+y^2 = 1,\ 0 \leq z \leq 1 \end{cases}$

(g) $\begin{cases} \vec{F}(x,y,z) = xy^2z(z-1)\vec{i} + x^2yz(z-1)\vec{j} + (z^3-z^2)\vec{k}, \\ S \text{ is the outer surface of the cylinder } x^2+y^2 = 1,\ 0 \leq z \leq 1 \end{cases}$

(h) $\begin{cases} \vec{F}(x,y,z) = \dfrac{x\vec{i} + y\vec{j} + z\vec{k}}{(x^2+y^2+z^2)^{3/2}}, \\ S : 4x^2+9y^2+6z^2 = 36, \text{ outward orientation.} \end{cases}$

(i) (20182-CK-3) $\begin{cases} \vec{F} = (x^2+y)\vec{i} + (x+2y)\vec{j} + (x+y+z)\vec{k}, \\ S : |x-y| + |x+2y| + |x+y+z| = 1, \text{ outward orientation.} \end{cases}$

(j) (20152-CK-3) $\begin{cases} \vec{F} = (x^3-z)\vec{i} - y\vec{j} + (3y^2z+2y)\vec{k}, \\ S : x^2+y^2+z^2 = 1, \text{ outward orientation.} \end{cases}$

(k) (20152-CK-7) $\begin{cases} \vec{F} = (6z^3-9y)\vec{i} + (3x-2z^3)\vec{j} + (3y-3x)\vec{k}, \\ S : x^2+3y^2+z^4 = 1,\ z \geq 0, \text{ oriented upward.} \end{cases}$

Exercise 90. Let $\vec{F} = x(y+z)\vec{i} + y(z+x)\vec{j} + z(x+y)\vec{k}$, and let L be the intersection of the cylinder $x^2+y^2+y = 0$ and the upper hemisphere $x^2+y^2+z^2 = 2,\ z \geq 0$. Prove that the circulation of $\vec{F}$ along L is equal to 0.

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